The Aerospace Dossier: Global UAS Certification Architecture

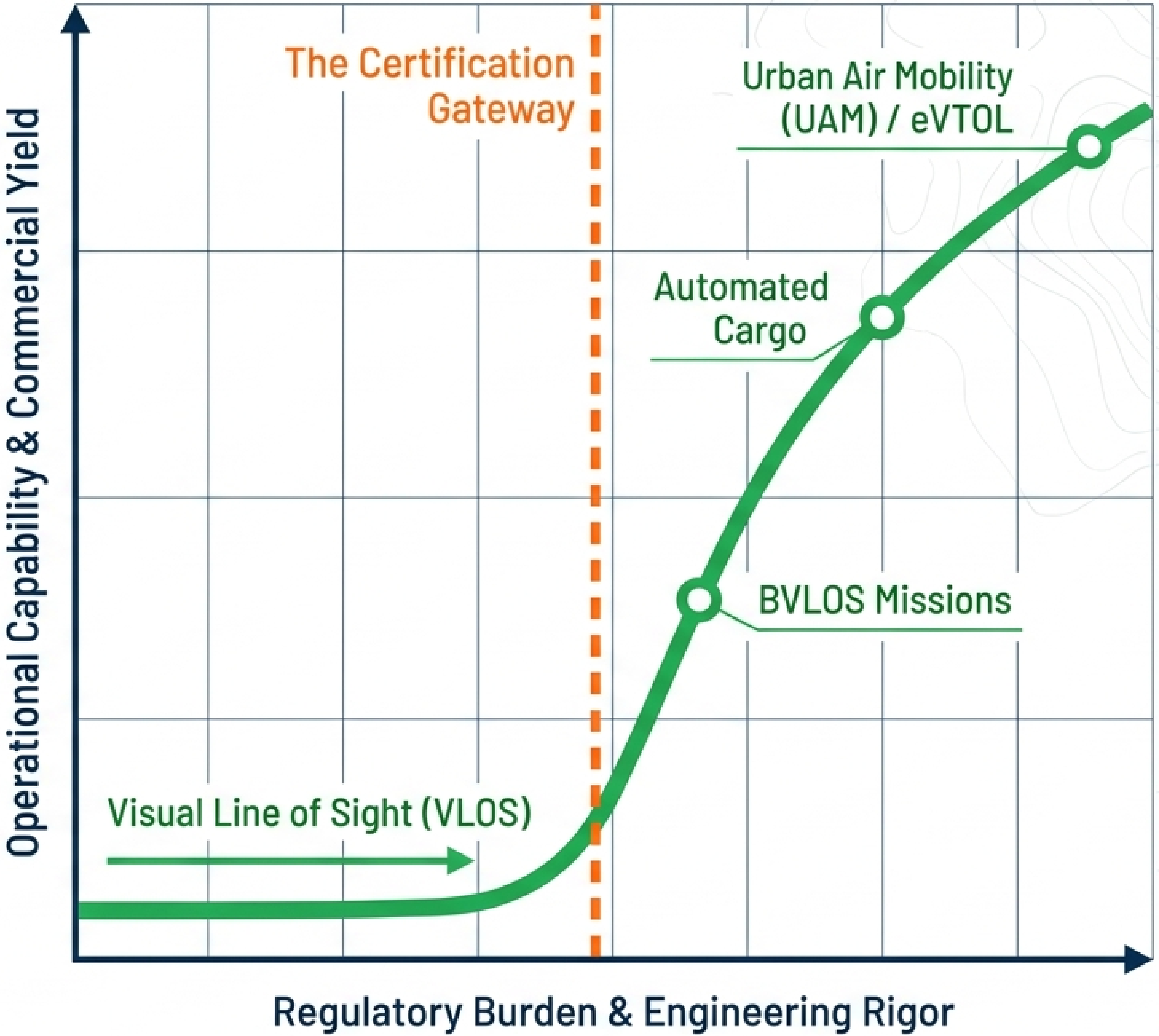
Modern drone certification is entirely governed by risk-based proportionality.

Moving from localized visual line-of-sight (VLOS) to scalable commercial autonomy requires crossing the threshold of comprehensive Type Certification.

The Design Requirement

Without Type Certification, operators are bound by rigid operational limits (e.g., <55 lbs, no flight over people).

Type Certification verifies that the aircraft design itself is robust enough to inherit the operational risk.



Transatlantic Risk-Tier Stratification

	EASA European Union Aviation Safety Agency	Federal Aviation Administration
Low Risk (Baseline)	**Open Category** • No prior authorization; A1/A2/A3 subcategories based on proximity to people; C0-C4 drone class identifications.	**Part 107** • Pilot certificate required, visual line of sight, <55 lbs. Focus is on the operator's knowledge.
Medium Risk (Mitigated)	**Specific Category** • Requires risk assessment and CAA authorization. Uses Standard Scenarios (STS) and PDRAs.	**Waivers & Exemptions** • Custom BVLOS approvals, specialized operational limits.
High Risk (Traditional)	**Certified Category** • Comprehensive certification of UAS, operator, and remote pilot for transporting people/dangerous goods.	**Type Certification (§21.17b) & Part 135** • Demonstrated airworthiness and design integrity for scalable UAM and delivery.

Structural Insight: EASA rules focus categorically on the operation type; FAA rules traditionally focus on operational boundaries and the pilot, bridging the gap with custom certification bases.

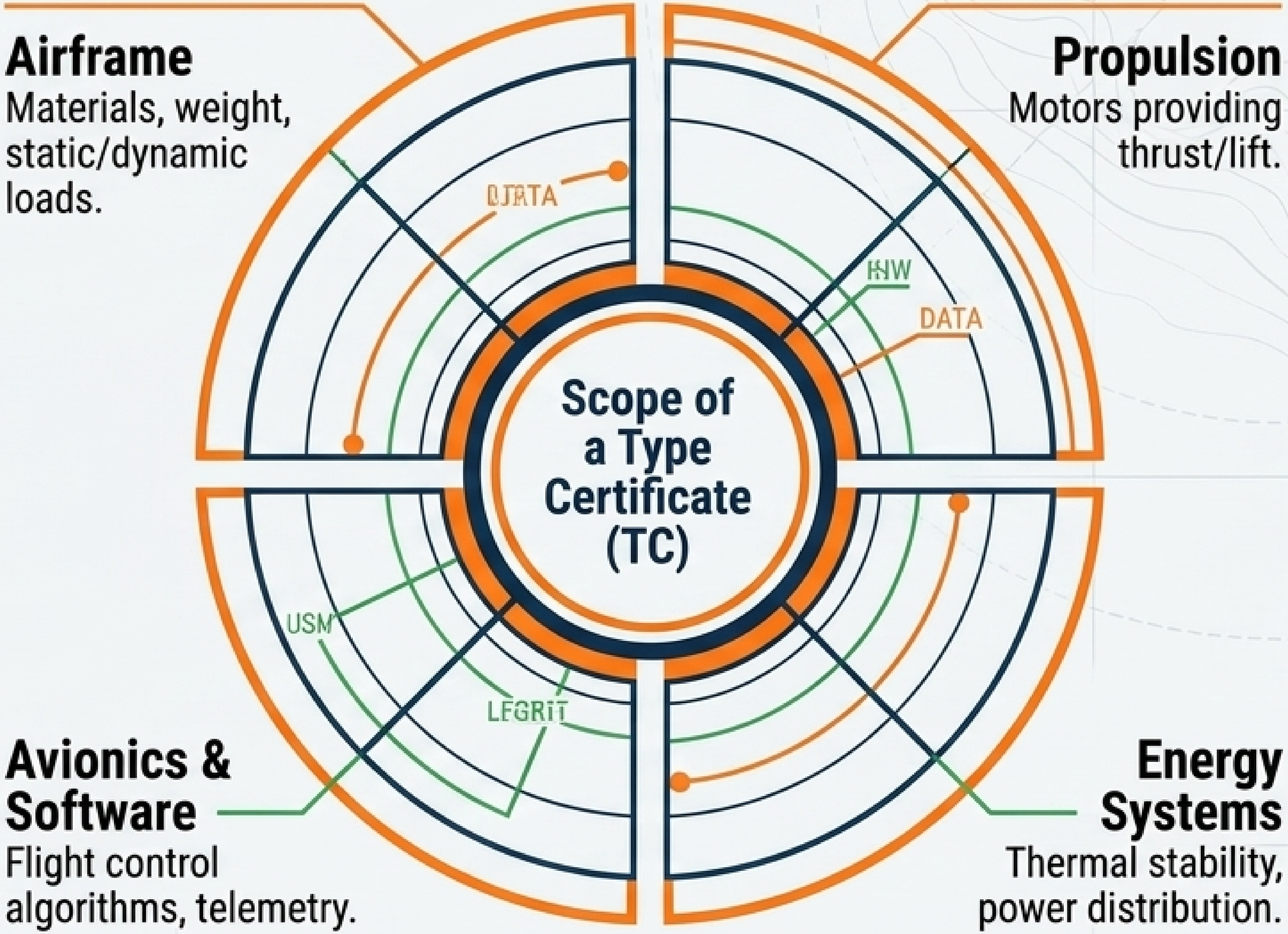
The Special Class Pathway (§21.17b)

The Problem

No comprehensive, universally accepted airworthiness standards exist for fully autonomous or BVLOS drones.

The Mechanism

The FAA uses 14 CFR §21.17(b) 'Special Class' pathways. This requires a custom Certification Basis for each UAS type, utilizing consensus standards (ASTM, RTCA).



Key Takeaway: A Type Certificate (TC) is the foundational approval. It signals the design is robust enough for complex, automated NAS operations, moving beyond mere operator knowledge.

SC-VTOL-01 Diagnostic Matrix

Applicability: Person-carrying Vertical Take-Off and Landing (VTOL) aircraft in the small category (Max certified take-off mass ≤ 2,000 kg; 5 seats or less).

Category Basic	Category Enhanced
Airworthiness Requirement Aircraft must be capable of a controlled emergency landing after a critical malfunction of thrust/lift.	**Airworthiness Requirement** Aircraft must be capable of continued safe flight and landing after a critical malfunction.
Operational Limitation Cannot be used for Commercial Air Transport (CAT) of passengers or operations over congested urban areas.	**Operational Allowance** Mandatory for CAT operations and flights over congested areas (cities/towns).

Regulatory Principle: SC-VTOL establishes performance-based safety and design objectives rather than prescriptive design solutions, utilizing AMC (Acceptable Means of Compliance).

The SORA 2.5 Data Funnel

Qualitative Inputs

Operational Concept, Intended Airspace, Hazard Identification

The Methodology

JARUS SORA 2.5 (Specific Operations Risk Assessment). Shifts assessment from qualitative assumptions to quantitative, data-backed conclusions.

Quantitative Outputs

Derivation of Safety Requirements, Risk Boundaries, Demonstration of Compliance

Standard Scenario (STS)

Pre-identified mitigation measures for common operations; operators declare compliance.

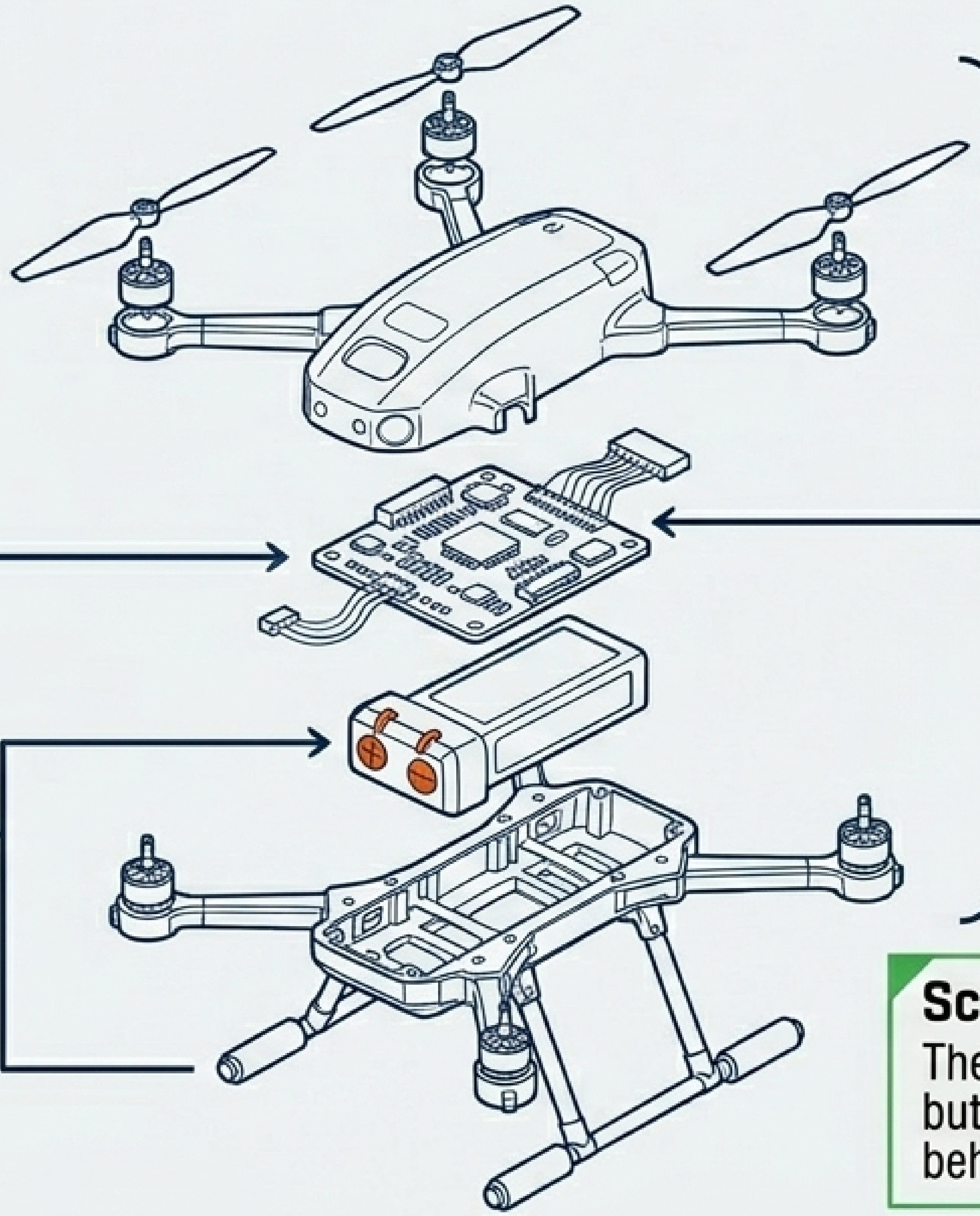
Pre-Defined Risk Assessment (PDRA)

Structured method for assessing medium-risk Specific category operations, streamlining CAA authorization.

Airworthiness: The Engineering Reality

Structural Testing
Verifying the airframe withstands static and dynamic loads (limit and ultimate thresholds).

Energy & Power
Assessing thermal runaway stability, charging logic, and fault detection mechanisms.



System Integration
Hardware-in-the-loop (HIL) simulation, validating loss-of-link protocols and navigation logic.

Environmental Testing
Evaluating operational performance under extreme temperature variations, humidity, vibration, and EMI/EMC interference.

Scientific Validation
The goal is not just proving the drone flies, but empirically validating fail-safe behaviors and extreme-condition resilience.

The D&R Execution Pipeline

The D&R Philosophy

A performance-based pathway. Rather than proving every nut and bolt meets a prescriptive standard, the manufacturer demonstrates aggregate safety through real-world operational validation.

ITCB → CONOPS → Cert Basis → Compliance → Testing → Conformity

ITCB

Initial TC Briefing: Alignment with regulators on configuration, propulsion, and intent.

CONOPS

Concept of Operations: Defining strict geographical boundaries and risk profiles.

Cert Basis

FAA publishes specific airworthiness criteria in the Federal Register.

Compliance

Mapping technical requirements to specific methods (HIL, chamber testing).

Testing

Ground & Flight Testing: Proving reliable operation across thousands of flight hours.

Conformity

Inspections guarantee the final production unit matches the documented design.

The Certification Triad Model

Type Certificate (TC)

The Design. Proves architecture and systems meet rigorous safety and airworthiness standards.

Production Certificate (PC)

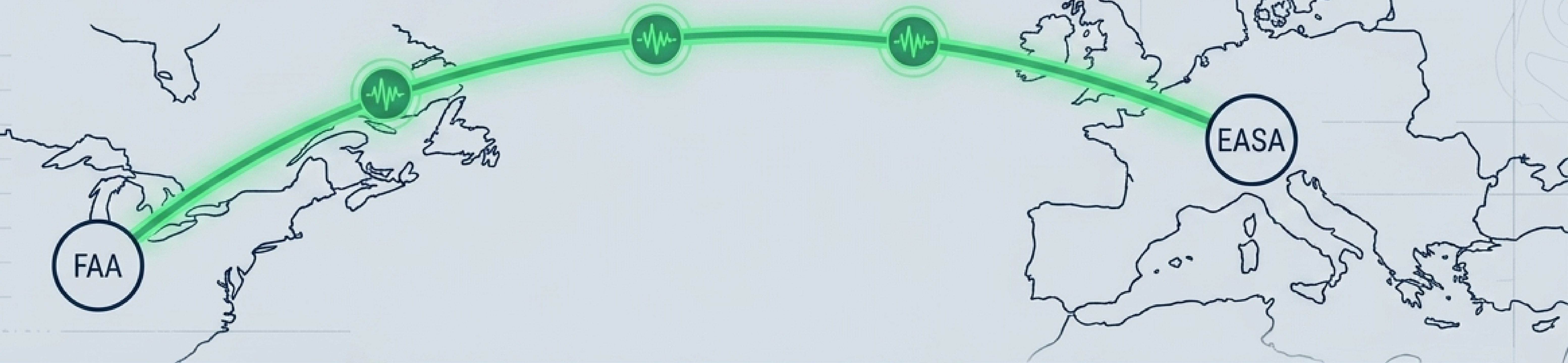
The Manufacturing. Proves the ability to repeatedly reproduce the TC-approved configuration with exact quality control.

Part 135 Certification

The Operation. The authorization required to use the certified aircraft in commercial air services for compensation.

A Type Certificate proves the drone is safe. A Production Certificate proves you can build it safely at scale. Part 135 proves your organization can operate it safely as a business. Achieving commercial scale requires all three gears to turn together.

Efficient Certification & Multilateral Validation



The CMT

Certification Management Team strengthening cooperation between major authorities (EASA, FAA, Transport Canada, ANAC) to handle technological surges.

Bilateral Procedures

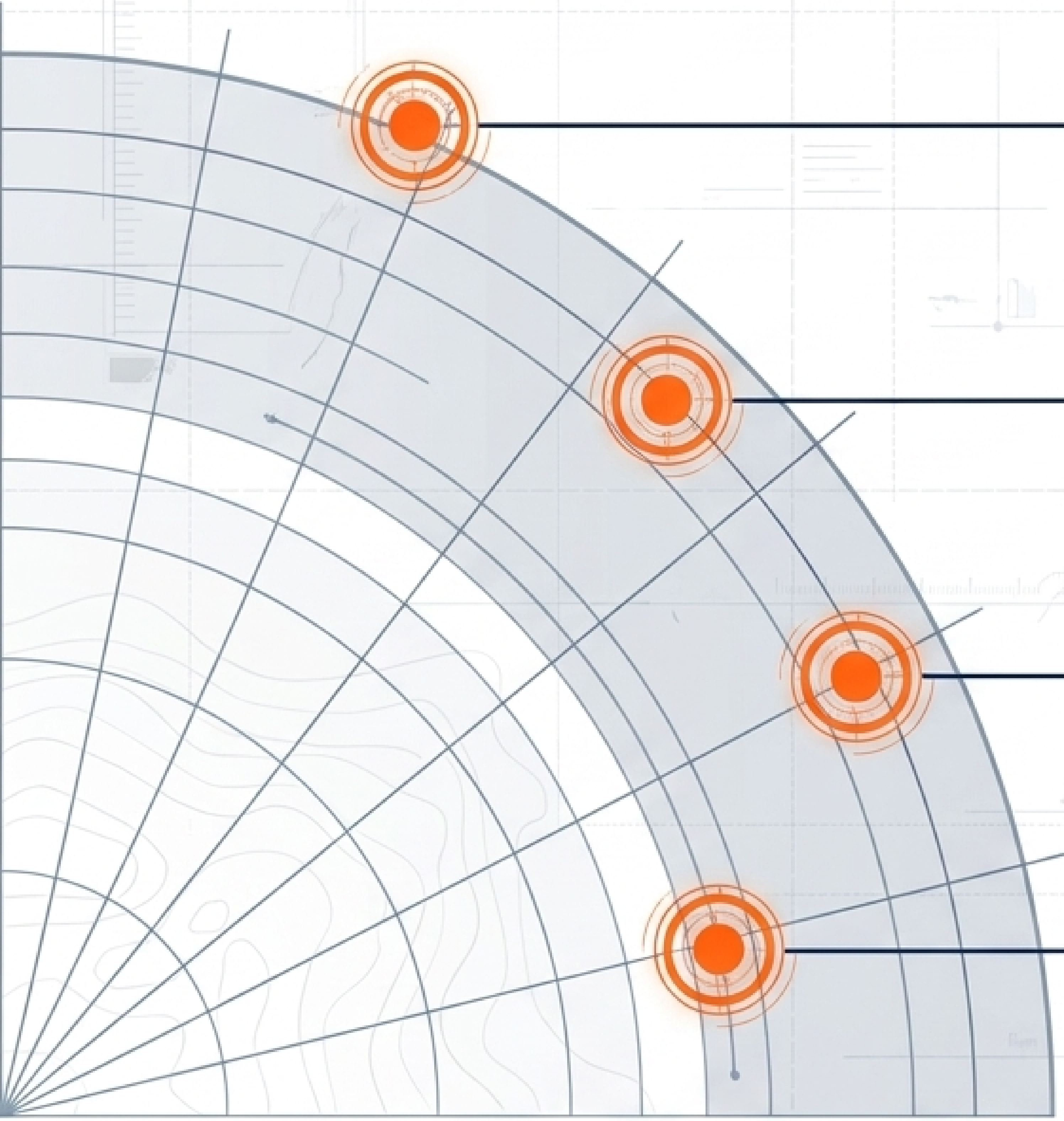
Revising implementation agreements to increase the scope of streamlined, mutual validation of certificates.

The Goal

A manufacturer achieving EASA certification should not have to repeat the engineering testing process for the FAA. Objective: interoperability and reduced duplication.

> “Complexity is growing fast... we cannot afford certification to take longer than before. The way we work today is not fit to manage the complexity of tomorrow.”

The Certification Horizon: 2025–2030



Digital Traceability
Shifting from paper trails to modern digital tracking of aviation parts globally to counter supply-chain threats.

Expanded SMS
Extending Safety Management Systems deeply into Design, Manufacturing, and Repair stations, beyond just operational limits.

Hybrid-Electric Propulsion
Developing novel protocols for energy storage and mitigating thermal runaway risks in integrated engines.

Smart Cockpits & Autonomy
Evaluating Human Factor integration with advanced AI, shifting focus away from traditional remote piloting.

AVIATE, NAVIGATE, COMMUNICATE: SAFETY IS NOT A DESTINATION, BUT A PRINCIPLE WE LIVE BY